

MMET 410 – Lab 7

Instructions

- I will demonstrate to you the pneumatic setup which is task 1 and task 2. They don't have any questions.
- In your report, include tasks 3A, 3B
- The format will be the same as previous labs
- Due date is tomorrow midnight

Task 1 and 2

- Air filter and air regulator
- **Single** and **double-acting** air cylinders
- Needle valve
- Directional control valves:
 - **Air-pilot**
 - **Manually operated**
 - **Solenoid control**

air-pilot controlled air cylinder (single and double acting)

manually controlled

solenoid

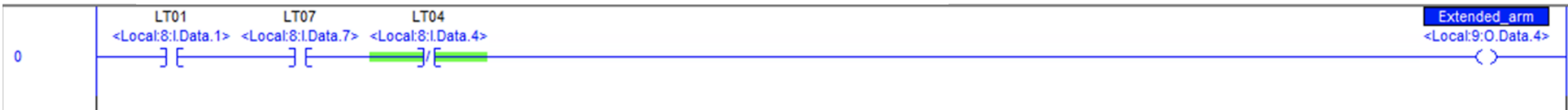
Task 3A

- Procedure:
push start_button ---> system on ---> LS1 triggered ---> solA on ---> LS2 triggered ---> solA off ---> stop_button ---> system off
- Inputs
start_button IN-0, stop_button IN-1
LS1 IN-2, LS2 IN-3
- Outputs
system OUT-0, SolenoidA OUT-1
- Sometimes the piston could be not fully extended (LS1 is off) and not fully retracted (LS2 is off), but the solA need to keep running.

Task 3B

- Write the program line-by-line by referring each row of the table.
- Same robot motion should be put in the same rung
- Each motion are abstracted to an PLC output.
- Each motion has its own prior on conditions and limit conditions.

Robot Motion	PLC Output	Limit Travel (PLC Input)	Prior <u>On</u> Conditions (PLC Inputs)
Extend arm	Local:9:O.Data04	04	01,07
Lower arm	Local:9:O.Data03	03	04,01,07
Grasp	Local:9:O.Data06	06	03,01
Lift arm	Local:9:O.Data02	02	06,01
Retract arm	Local:9:O.Data05	05	02,06,01
Rotate CCW	Local:9:O.Data00	00	05,06
Extend arm	Local:9:O.Data04	04	00,06
Lower arm	Local:9:O.Data03	03	04,00,06
Open	Local:9:O.Data07	07	03,00
Lift arm	Local:9:O.Data02	02	07,00,04
Retract arm	Local:9:O.Data05	05	02,00,07
Rotate CW	Local:9:O.Data01	01	05,00,07

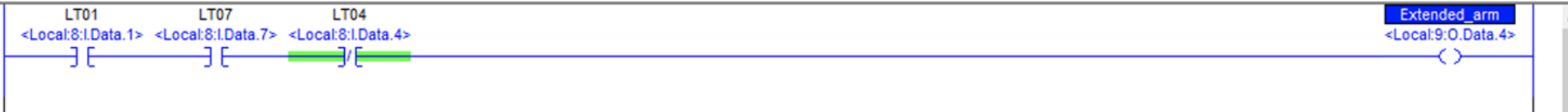


Task 3A

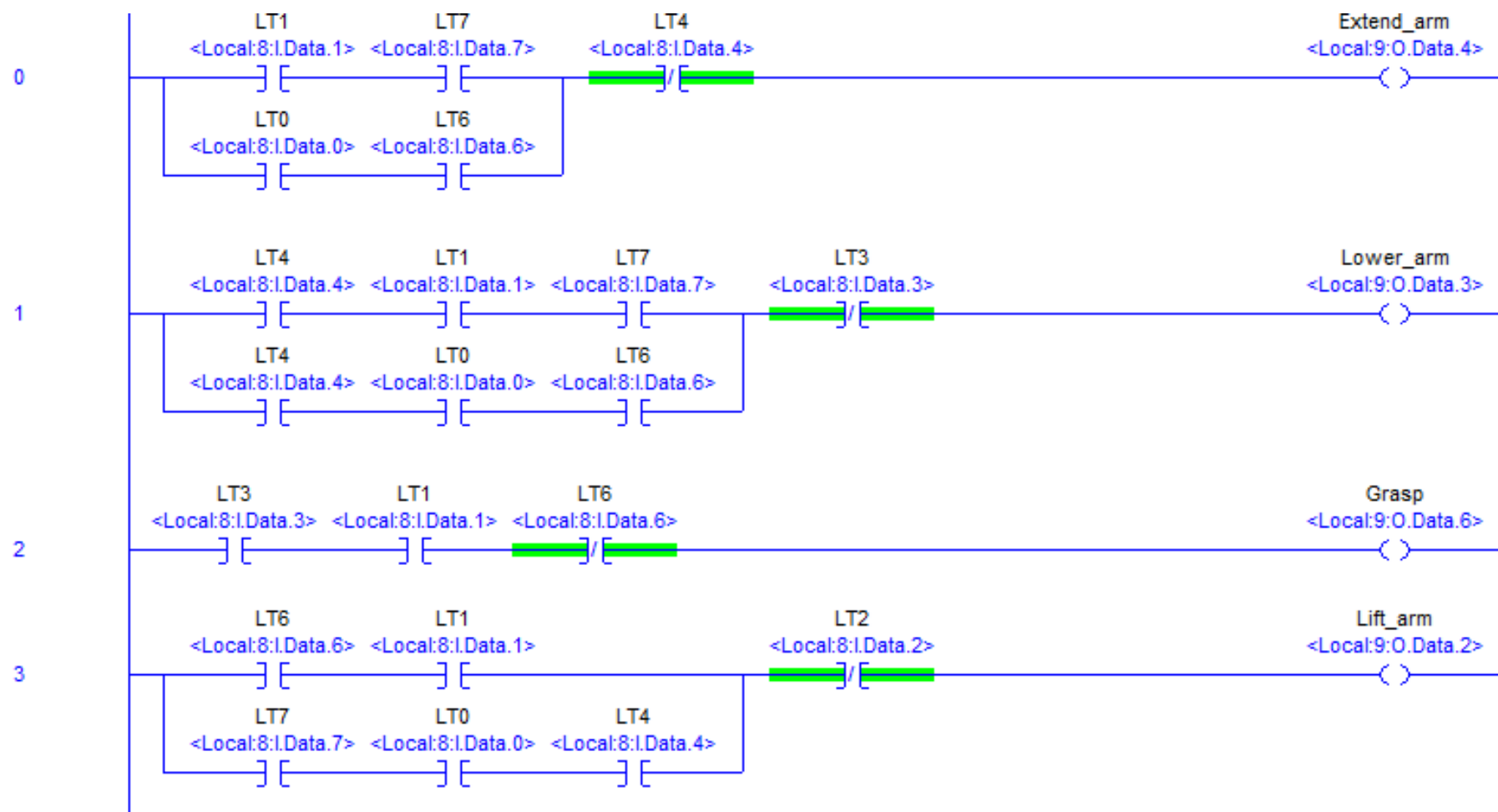
- Inputs
start_button IN-0, stop_button IN-1
LS1 IN-2, LS2 IN-3
- Outputs
system OUT-0, SolenoidA OUT-1
- Procedure:
push start_button ---> system on -
--> LS1 triggered ---> solA on ---
>LS2 triggered ---> solA off --->
stop_button ---> system off
- Sometimes the piston could be not fully extended (LS1 is off) and not fully retracted (LS2 is off), but the solA need to keep running.

Task 3B

- You don't need to write down entire action sequence, could simply copy from document or just write "follow the table"
- Write the program line-by-line by referring each row of the table.
- Same robot motion should be put in the same rung
- Each motion are abstracted to an PLC output.
- Each motion has its own prior on conditions and limit conditions.



Task 3B



Task 3B

